FIRST SEMESTER 2026-2027
COURSE HANDOUT

Date: 03.08.2026

In addition to part I (General Handout for all courses appended to the Timetable) this portion gives further specific details regarding the course.

Course number: CE F230

Course title: Civil Engineering Materials

Instructor-in-charge: Dr. Mukund Lahoti

Team of Instructors:

Lectures: Mukund Lahoti **(1232)**
Tutorials: Mukund Lahoti **(T1-6103)**, Shubham Sharma, Smriti Mishra **(T2-6104)**
Practical:P1: Daltej Singh (RS), Rishabh Sharma (RS), Kumar Rishabh (RS), Rohit Narayan (RS), Deina Susan Sony (RS) , **P2:** Vignesh Ganesan (RS), Daltej Singh (RS), Suresh S (RS), Shubham Sharma (RS), Rohit Narayan (RS) **(Concrete Lab 1124)**

1. **Course Description:**

Different types of cements, chemical composition, properties and tests, coarse and fine aggregate for concrete, tests on aggregates, grading of aggregates and its effect on concrete properties, chemical and mineral admixtures, properties and tests on fresh and hardened concrete; transportation and placing of concrete, non-destructive testing of concrete, durability of concrete, quality control and acceptance criteria of concrete, Factors in the choice of mix proportions, Proportioning of concrete mixes by various methods – BIS method of mix design; Introduction to special concretes. Manufacturing/sources, classification, applications, properties and testing of bricks, blocks, tiles, aggregates, lime, timber, paints, glass, bitumen, cutback, emulsion, modified bitumen, steel, non-ferrous metals, polymeric materials, geo-synthetics, etc. Low cost and waste material in construction. Latest, BIS, IRC & ASTM specifications and guidelines of all above-mentioned material, and construction equipment.

2. **Scope and Objective:**

The course deals with properties, characterization, testing, selection and quality control of construction material. For economical, durable and high performance of structure, it is imperative to have knowledge on construction materials and their wider applications. This course provides a basic and enhanced overview of various construction materials presently used in practice.

3. **Course learning outcomes:**

By the end of this course, students will be able to:

a) **Identify and classify common civil engineering materials** such as concrete, steel, wood, aggregates, asphalt, and modern composites.

b) **Explain the physical, mechanical, and chemical properties** of construction materials and how these properties affect material performance and selection.

c) **Perform standard laboratory tests** to evaluate the properties and behaviour of civil engineering materials, including tests for strength, and workability.

d) **Interpret test data and analyse results** to assess material suitability for specific applications in civil engineering.

e) **Understand the influence of material properties** on the structural performance, serviceability, and durability of civil infrastructure.

f) **Compare and evaluate materials** based on environmental impact, cost-effectiveness, availability, and sustainability.

g) **Demonstrate awareness of current trends and innovations** in civil engineering materials, including smart materials, green materials, and recycled materials.

h) **Apply appropriate material selection** in the design and construction of civil engineering structures in accordance with relevant codes and standards.

4. **Textbooks and references**:

Textbooks:

T1. Gambhir, M.L. (2013) "Concrete Technology" Tata McGraw-Hill Publishing Company Ltd, 5th edition.
T2. Duggal, S.K. (2012) "Building Materials" New Age International Pvt. Ltd., 4th edition.
T3. Gambhir, ML and Jamwal, N. (2014) "Lab Manual- Building and Construction Materials (Testing and quality control)", Mc Graw Hill, 1st edition **(Lab Manual)**

Reference Books:

R1. Relevant B.I.S, IRC, ASTM codes
R2. Mehta, P.K. and P.J.M. Monteiro. (2006) "Concrete Microstructure, Properties, and Materials", The McGraw-Hill Companies, United States.
R3. Neville, A.M. (1995) "Properties of Concrete", Pearson Education, 4th edition
R4. Gambhir, M.L., and Jamwal, N. (2011) "Building Materials- Products, Properties and Systems", Mc Graw Hill, 1st edition
R5. Moondra, HS and Gupta, R. (2000) "Lab Manual for Civil Engineering", CBS, 2nd edition **(Lab Manual)**

5. **The course will be conducted through in-person lectures (M, W 4, T 10; 1232), tutorials (F 9; T1 6103, T2 6104), and practical (P1: Th 7 8; P2: T 7 8).**

6. **Course Plan:**

Module	Lecture Session	Lec no.	Ref.	Learning outcome
1	Civil engineering materials- An overview	1-6	Slides	Understand various construction materials, the importance of concrete, process of cement
	Concrete as a construction material		1, T1	

	Cement and hydration, water/cement ratio and supplementary cementitious materials		2, T1 and 5, T2	manufacturing, cement hydration, microstructures, and SCMs
2	Aggregate classification and tests	7-10	3, T1 and 6, T2	Understand important aggregates sources, their classification, influence of properties of aggregates on their performance, testing methods/standards
3	Chemical admixtures	11	5, T1	Develop clear understanding of the purpose and mechanism of chemical admixtures
4	Concrete mix design	12-15	IS10262, 10, T1	Design concrete mix and explain the effect of concrete composition on concrete fresh properties, transportation and handling of concrete
	Fresh Concrete	16-17	6, 7 T1	
	Working with fresh concrete	18	11, 12, T1	
5	Hardened concrete properties	19-21	8, T1	Explain the effect of composition on hardened concrete properties and various concrete tests including NDTs
	Testing on hardened concrete	22-23	8, 13 T1	
6	Durability of concrete	24-26	15, T1 and slides	Examine various durability issues in concrete and recommend preventive measures
7	Special Concretes	27-28	14, 16, T1	Categorize special concretes and understand features of their mix design
8	Clay products: Properties, tests, selection	29-30	2, T2	Describe important ingredients, manufacturing and classification of bricks. Define important terms in brick masonry. Experiment key properties of bricks.
9	Wood and timber	31-33	4, T2	Indicate salient parts of timber in X-section of wood. List defects. Applications.
10	Steel, ferrous and non-ferrous metals	34-35	14, 15, T2	Describe the manufacturing of steel, classification, applications. Prepare a comprehensive list of important sources, properties and usage of aluminum, copper, zinc, lead, tin and nickel.

11	Lime: Properties, tests, application	36	8, T1	List steps involved in lime manufacture, impurities & effects, lime properties, tests and applications.
12	Introduction to highway materials	37	19, T2, Relevant IS and IRC codes	Understand concisely various materials involved in highway construction
13	Polymeric material	38-39	21, T2, Slides	Understand important properties and applications of FRPs, and FRCs
14	Geo-synthetics and Construction equipment, revision	40-42	21, T2, Slides	Understand important properties and applications of geosynthetics. Glimpse of significant construction equipment.

5. Evaluation Scheme:

Component	**Duration**	**Weightage (%)**	**Date & Time**	**Nature of component (Close Book/ Open Book)**
Class assessment	5 min	8	Lecture hour	OB
Tutorials	30 min	12	Tutorial hour	CB
Mid-semester exam	90 min	25	05/10 AN1	CB
Laboratory	-	20	Lab record (10%) Lab quiz (10%)	OB + CB
Comprehensive Examination	180 min	35	03/12 FN	CB + OB

7. Consultation Hour: Tuesday 4-5 pm. Please email at mukund.lahoti@pilani.bits-pilani.ac.in if needed.

8. Notices: Announcements and course material will be communicated via Nalanda.

9. Make-up Policy:

a. Take prior permission from Instructor in charge.
b. Make-up will be granted on a case-by-case basis only for genuine reasons.
c. For medical cases, a certificate from the physician concerned of the Medical Centre must be produced.
d. No make-up for tutorials (Best 6 of ‘n’ will be considered).
e. One make-up lab.

10. Note: If you miss a complete component (Lab/Tutorial/class-assessment/mid-semester exam/comprehensive exam), you may be granted NC.

Instructor-in-charge
Course No. CE F230

Civil Engineering Materials (Laboratory)

Instructor-in-charge : Dr. Mukund Lahoti

Textbooks:

1. Gambhir, ML and Neha Jamwal, Lab Manual- Building and Construction Materials (Testing and quality control), Mc Graw Hill, 1st edition 2014
2. Moondra, HS and Rajiv Gupta, Lab Manual for Civil Engineering, CBS 2nd edition 2000

LIST OF EXPERIMENTS

Week	Name of experiment
1	Introduction to all experiments, marking scheme, lab report format, etc.
2	**(a)** Heat of hydration (b) Rebar testing
3	Cement: (a) Normal consistency and setting time (b) Fineness of cement
4	Fine aggregate: (a) Bulking of sand (b) Silt content (c) Specific gravity
5	Fineness modulus and grain size distribution: Fineness modulus and grain size distribution
6	Cement mortar: ○ Cast 3 cubes for compressive strength testing (air dried in shade, water-cured) ○ Cast 3 briquettes for tensile strength testing
7	Concrete (casting): (a) Nominal mix design ○ Cast 2 cylinders, 2 cubes, 1 beam ○ Test workability (slump)
8	Concrete (casting): IS 10262 mix design ○ Cast 2 cylinders, 2 cubes, 1 beam ○ Test workability (slump)
9	Concrete testing: (a) Modulus of rupture (beam test) (b) Test compressive strength of cube and cylinder (c) Split tensile strength of cylinder (d) RCPT (e) NDT
10	Brick: a) Compressive Strength b) Casting of Fly ash Brick
11	Make-up lab
12	Lab quiz